Hiraia — progress update

Early September 2026

An offline, on-device science tutor for Filipino grade-school students (Grades 3–10), aligned to the DepEd MATATAG curriculum, in Tagalog, Cebuano and English. The published build (v0.1) targets Android 12+ phones with 6 GB+ memory; a 1B budget tier runs on a Redmi (Snapdragon 685) in unpublished builds.

36,487
illustrated fact cards

50,279
verified trilingual facts

324
MATATAG competencies tracked (321 with cards)

25,746
cards with a quiz question

Evidence: repository at commit 58d358495 (branch question-cards) and its evaluation records; the published app is app v0.1 — the feed shown here ships in the next build. Sources are in the slide notes.

hi Training runs on RunPod

Most of the compute went into a continued-pretrained Filipino base model; the rest into tutoring fine-tunes, a pruning experiment, and evaluation.

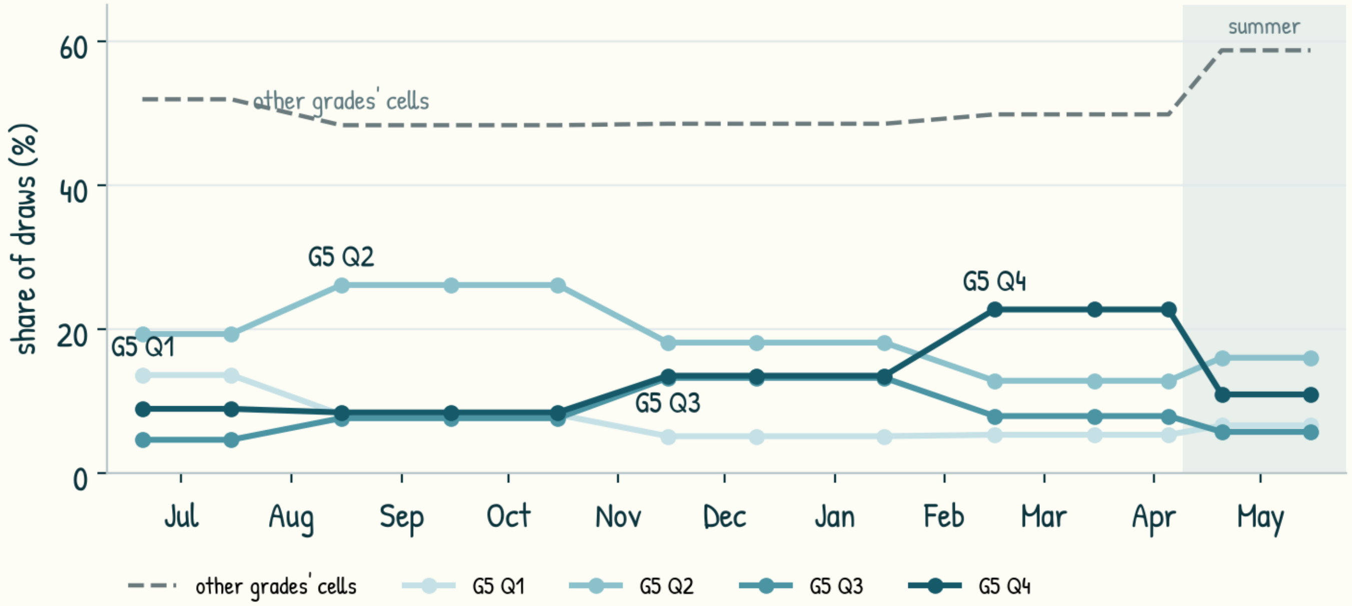
run	GPUs	GPU-h	cost	basis	outcome
Full continued pre-training — Qwen3.5-2B, ≈13.7B tokens	8 × H100 80GB	294	$972	cost documented; hours from pod age	Held-out perplexity: Tagalog −80%, Cebuano −82%, English −4% (improved)
CPT probe run 2 (full-parameter, ZeRO-2)	8 × H100	167	$550	documented	Only checkpoint-125 (525M tokens) survived a cross-DC sync failure — it validated the thesis: perplexity −63% tl / −63% ceb; GGUF export de-risked
CPT probe run 1 (≈5.1B tokens, trained as LoRA by mistake)	8 × H100	85	$285	documented	No usable artefact; the pod/volume/heartbeat pipeline was proven
Full CPT attempt 1 (aborted)	8 × H100	66	$216	estimate	Pre-processing collapse (worker count vs cgroup quota); run held and restarted
Prune + heal 2.4B experiment (Sailor2-3B)	8 × A100	56	$86	documented	Heal recovered fluency (ppl tl 6.7 / ceb 9.9); conservative re-SFT under-fit — not shipped
GRPO rounds 1–5 (1B/3B)	1–4 × H100 / RTX 6000	55	$74	mostly documented	Not shippable: capability 3.30 ⟶ 2.98; English +1.4 but abstention and Bisaya regressed
SFT v1 / v2 on the CPT base (Qwen3.5-2B)	1 × A100 80GB	3	$6	documented	v1 is the ship candidate: reply-language accuracy tl 99% · ceb 99% · en 94%
Adapter iterations on Sailor2-3B / 1B and tooling (36 entries, May–June)	H100 / L40 / A100 / L4	60	$128	mostly estimated	Shipped adapters v3 ⟶ v11: grounding faithfulness, static-prompt TTFT, kitten/cat tiers; QVAC-native training path evaluated and dropped
Checkpoint-sync, eval and debug pods	various	68	$85	mixed	Evaluation on a device-equivalent GGUF; pre-flight debugging of run 2

≈853 GPU-hours and ≈$2,402 on RunPod since late May 2026 (rows sum to the totals); ≈$2,000 of that from figures recorded at the time, the rest estimated at list price — no billing export yet. About 80% of the compute went into continued pre-training of the Filipino base model.

A curriculum-aware feed

Cards are weighted by the student's grade and the curriculum quarter inferred from the device date; the boost moves through the school year by itself.

Grade-5 student: share of feed draws by cell, SY 2026-27 (untagged cards ≈1.5%, not shown)



current quarter, same grade	adjacent quarter	other quarter, same grade	adjacent grade, current	off-curriculum
×6	×3	×1.5	×2	×0.4

placeholder — screenshot of the unpublished question-cards build (commit 58d358495); not in the published v0.1 APK

fact card

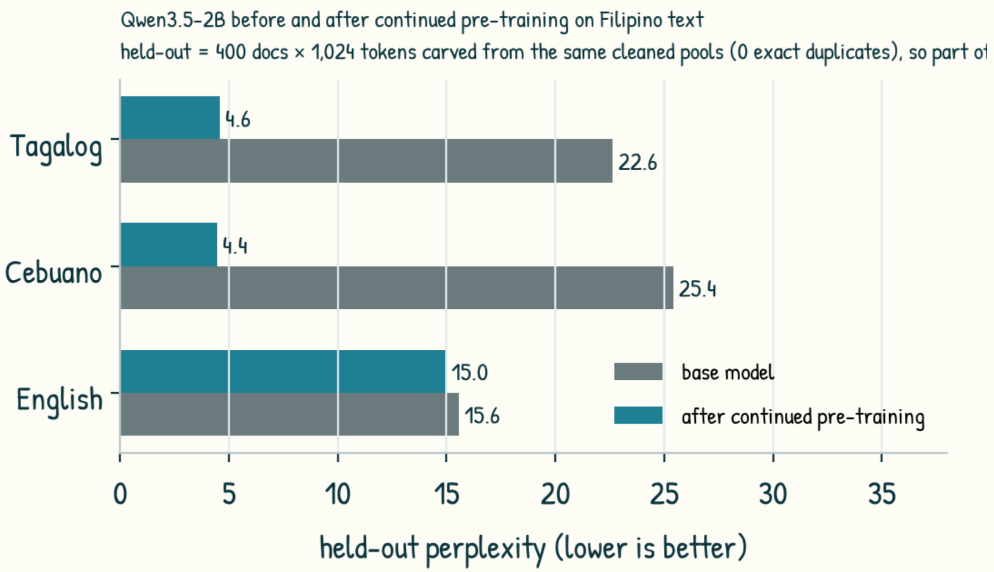
placeholder — screenshot of the unpublished question-cards build (commit 58d358495); not in the published v0.1 APK

interject quiz card

- 36,487 illustrated cards; 34,719 facts labelled to 324 competencies (LLM labels, 83% inter-model agreement; single-model cells down-weighted)
- 25,746 cards (70%) carry a verified quiz question
- 0 elementary competencies below floor; 38 junior-high competencies below floor in the card pool (16 once DepEd module cards are counted)

Model progress: fluency and pedagogy

From a fluent-but-cautious 3B adapter line to a continued-pretrained 2B base that reads and writes Tagalog and Cebuano: fluency is measured on held-out text, pedagogy on scripted tutoring probes.



Tiles: pedagogy figures are from the June 3B line; language accuracy and hedging from the August 2B (SFT v1). The 2B capability run is pending.

3.49 ⟶ 3.72

capability score (0–5), 102 tutoring probes — June 3B line; data rebalance alone, hybrid retrieval

99 · 96 · 94%

reply-language accuracy, SFT v1 2B (Tagalog · Cebuano · English); Cebuano 99% after correcting fastText tl/ceb mislabels

64%

of Tagalog replies open with an 'I'm not sure' hedge (SFT v1 2B); about two-thirds go on to explain the term anyway

Pedagogy (measured on scripted probes)

Scripted probes (102 in June, 143 now; 11 tiers, LLM-judged 0–5) showed the June model fluent but over-cautious: 19 of 91 answerable questions were refused. A retrieval fix plus a small data rebalance (+82 rows) lifted the score 3.32 ⟶ 3.72 (rebalance alone +0.22), and grounding faithfulness was fixed — the tutor defers to retrieved verified facts and abstains when none apply. The 2B line inherits the same harness; its capability run is the next measurement.

Known weaknesses (measured)

- Hedging: the SFT v1 2B prefixes 64% of Tagalog replies with a 'not sure' clause on grade-5 terms; about two-thirds of those then explain the term — a template-wording effect, not a knowledge gap
- English mode-lock is the 2B's fragile edge (94% vs 99% in Tagalog and Cebuano); misroutes fall to Tagalog
- The 1B budget tier cannot learn safety negation (DPO tested and rejected); junior-high content is thinner than elementary

placeholder — screenshot of the unpublished question-cards build (commit 58d358495); not in the published v0.1 APK

grade + language settings

hi Next 2–3 weeks

Consolidate the model, then prove it on the device.

- More SFT and knowledge distillation on the CPT base (mode-locked buckets, teacher-generated tutoring turns)
- Extensive synthetic testing on-device: scripted student sessions on the Redmi, TTFT and per-page latency budgets
- Model-as-judge quality control on every generated fact, quiz and translation (decorrelated judge, AUP-safe routing)
- APK-level optimisations: bundle size (engravings subset, indexed PNG), cold start, SQLite migrations, edge-walk cost
- Native-speaker review of the Tagalog and Cebuano copy (onboarding, settings, hedging clause)
- Junior-high gap fill: Lane B facts for the 16 remaining thin competencies; module-card supply
- Finish the quiz lane (10.7k pool cards still without a question) and per-competency Miss-Card labels
- Over-the-air fact-bank updates so content ships without an APK
- Retrieval-QA cases per new competency before each gate run
- Draft a classroom-pilot protocol (consent, offline logging, pre/post measures) to propose to DepEd schools

To report next time: the 2B capability run on the same 143 probes; on-device TTFT and per-page latency on the Redmi; quiz coverage after the lane completes; bundle size after optimisation.

Links

Code, build and evaluation records are public; model weights follow once the release checklist is done.

Website	https://hiraia.org
APK (direct download)	https://hiraia.org/models/hiraia.apk v0.1.0, 325 MB, sha256 708ea605..., Android 12+, 6 GB+ RAM; mirror: github.com/helloluis/hiraia/releases/tag/v0.1.0
GitHub	https://github.com/helloluis/hiraia source, pipelines and evaluation harness (public)
Hugging Face	to be published model weights + evaluation kit: release pending; corpus provenance will be documented, DepEd-derived text is not redistributed